CASCADIA ANALYTICS

# Alphabet Inc.

Equity Research Note | GOOGL | 21 OCTOBER 2026

## The business is exceptional but the price discounts perfection

Alphabet Inc. screens as a Hold, Valuation Concerns with a $142.04 blended target, 58% below the current price.

**Analyst: Justin Wen**

| REFERENCE PRICE | BLENDED TARGET | BLENDED UPSIDE | |
|---|---|---|---|
| $340.27 | $142.04 | -58% | HOLD, VALUATION CONCERNS |

### Investment Thesis

The business is exceptional but the price discounts perfection. Alphabet owns the two most durable franchises in technology, Search and YouTube, plus the fastest-scaling cloud business in the industry. Our framework rates the business Attractive on quality, growth, and trend, yet the DCF puts fair value far below the market price. This is a great company at a demanding price. The capex supercycle is the crux of the valuation debate. Management guided 2026 capex to $195-205B and signaled a significant increase

- The business is exceptional but the price discounts perfection
- The capex supercycle is the crux of the valuation debate
- Cloud is the growth engine, Search is the cash engine under watch
- Valuation verdict: Hold, Valuation Concerns

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## The latest print

**The capex supercycle is the crux of the valuation debate.** Management guided 2026 capex to $195-205B and signaled a significant increase again in 2027. Q2 free cash flow turned negative for the first time as $44.9B of quarterly capex exceeded $39.1B of operating cash flow. The bull case requires this spend to convert into durable Cloud and AI revenue at high incremental margins; the bear case is a multi-year ROI shortfall.

**Cloud is the growth engine, Search is the cash engine under watch.** Google Cloud revenue grew 82% YoY to $24.8B with operating income tripling to $8.8B and a $514B contracted backlog. Search keeps compounding, but Meta's Muse agent (2.8M downloads in twelve days) and court-ordered data-sharing remedies are the first credible threats to the ad moat in two decades.

**Valuation verdict: Hold, Valuation Concerns.** Our probability-blended DCF target is $142.04, implying 58% downside from $340.27. Per our rating rules, an overvalued but healthy business is a Hold on valuation concerns, not a Sell. We would turn constructive on a capex normalization signal, a Search monetization re-acceleration, or a materially lower entry price.

| $120.0B | 34.0% | $24.8B | $514B | $195-205B | -$5.9B |
|---|---|---|---|---|---|
| Q2 2026 REVENUE | Q2 2026 OPERATING MARGIN | GOOGLE CLOUD Q2 REVENUE | CLOUD BACKLOG | 2026 CAPEX GUIDANCE | Q2 2026 FREE CASH FLOW |

## Key metrics in detail

| METRIC | LATEST | VS YEAR AGO | TREND |
|---|---|---|---|
| **Q2 2026 Revenue** | $120.0B | +24.4% YoY | Accelerating |
| **Q2 2026 Operating Margin** | 34.0% | +1.9pp | Expanding |
| **Google Cloud Q2 Revenue** | $24.8B | +82% YoY | Hypergrowth |
| **Cloud Backlog** | $514B | ~2x | Surging |
| **2026 Capex Guidance** | $195-205B | +119% vs 2025 | Supercycle |
| **Q2 2026 Free Cash Flow** | -$5.9B | First negative | Pressured |

## The moat

Google Cloud hypergrowth: +82% YoY revenue, tripled operating income, $514B contracted backlog converting over 24 months.

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Alphabet Inc. valuation framework

## The growth engine

Gemini scale: 950M MAU, 22B API tokens per minute, deepening enterprise AI adoption on Google infrastructure.

## Three scenarios make the valuation range explicit

Three scenarios, weighted 25% / 50% / 25% into the blended target.

| SCENARIO | WEIGHT | 2027-31 CAGR | 2031 REVENUE | 2031 EBIT MARGIN | TARGET | UPSIDE |
|---|---|---|---|---|---|---|
| **Bear** | 25% | 8.0% | $719.97B | 33.5% | **$61.12** | -82% |
| **Base** | 50% | 12.0% | $863.55B | 36.5% | **$131.29** | -61% |
| **Bull** | 25% | 16.0% | $1,029.17B | 40.0% | **$244.45** | -28% |

**Bear:** downside case at $61.12 (-82%).

**Base:** central case at $131.29 (-61%).

**Bull:** upside case at $244.45 (-28%).

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# The probability-blended target implies 58% downside

A 25% Bear / 50% Base / 25% Bull weighting implies -58% from the $340.27 reference price.

**$142.04**
BLENDED TARGET

The valuation uses a multi-year unlevered free cash flow DCF. The discount rate is the live CAPM rate: the 10-year Treasury yield plus beta times the market-implied equity risk premium. Terminal value is a Gordon perpetuity on terminal-year free cash flow.

Mechanics are explicit. The bridge runs revenue to EBIT to cash taxes, then adds back D&A and subtracts capex and working-capital investment to reach unlevered free cash flow. Targets are divided by diluted shares outstanding.

| | |
|---|---|
| Valuation horizon | **2027-2031** |
| Live WACC | **9.45%** |
| WACC path (Year 1 to final year) | **9.45% to 8.88%** |
| Terminal growth | **2.0% / 2.5% / 3.0%** |
| Net debt | **$56.90B** |
| 2031 FCF (Base) | **$148.43B** |
| 10-year Treasury yield | **5.1%** |
| Beta | **1.21** |
| Equity risk premium | **3.7%** |

**HOLD, VALUATION CONCERNS | $142.04 BLENDED TARGET | WEN**

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## The reverse DCF: what the market prices in

Holding the base-case FCF margin flat, the market-implied revenue CAGR is 62.5%. at/above Bull (16%) - euphoric

## Reverse DCF sensitivity

Implied enterprise value and share price at each five-year revenue CAGR, with the base-case free cash flow margin held flat. The highlighted row sits nearest the market-implied CAGR.

| CAGR | Implied EV | Implied price | CAGR | Implied EV | Implied price | CAGR | Implied EV | Implied price |
|---|---|---|---|---|---|---|---|---|
| -20% | $168.6B | $9.07 | +0% | $432.2B | $30.49 | +20% | $986.9B | $75.56 |
| -19% | $177.2B | $9.77 | +1% | $451.6B | $32.07 | +21% | $1025.6B | $78.70 |
| -18% | $186.2B | $10.50 | +2% | $471.8B | $33.71 | +22% | $1065.5B | $81.94 |
| -17% | $195.6B | $11.27 | +3% | $492.7B | $35.41 | +23% | $1106.7B | $85.29 |
| -16% | $205.4B | $12.06 | +4% | $514.4B | $37.17 | +24% | $1149.2B | $88.74 |
| -15% | $215.7B | $12.90 | +5% | $536.9B | $39.00 | +25% | $1193.1B | $92.30 |
| -14% | $226.3B | $13.77 | +6% | $560.2B | $40.89 | +26% | $1238.3B | $95.98 |
| -13% | $237.5B | $14.67 | +7% | $584.4B | $42.86 | +27% | $1284.9B | $99.77 |
| -12% | $249.1B | $15.62 | +8% | $609.5B | $44.89 | +28% | $1333.0B | $103.67 |
| -11% | $261.3B | $16.60 | +9% | $635.4B | $47.00 | +29% | $1382.5B | $107.69 |
| -10% | $273.9B | $17.63 | +10% | $662.2B | $49.18 | +30% | $1433.5B | $111.84 |
| -9% | $287.1B | $18.70 | +11% | $690.0B | $51.44 | +31% | $1486.1B | $116.11 |
| -8% | $300.8B | $19.81 | +12% | $718.8B | $53.77 | +32% | $1540.2B | $120.51 |
| -7% | $315.1B | $20.97 | +13% | $748.6B | $56.19 | +33% | $1596.0B | $125.04 |
| -6% | $329.9B | $22.18 | +14% | $779.3B | $58.69 | +34% | $1653.4B | $129.70 |
| -5% | $345.4B | $23.43 | +15% | $811.1B | $61.28 | +35% | $1712.5B | $134.50 |
| -4% | $361.4B | $24.74 | +16% | $844.0B | $63.95 | +36% | $1773.2B | $139.44 |
| -3% | $378.1B | $26.10 | +17% | $878.0B | $66.71 | +37% | $1835.8B | $144.52 |
| -2% | $395.5B | $27.51 | +18% | $913.2B | $69.56 | **+38%** | **$1900.1B** | **$149.75** |
| -1% | $413.5B | $28.97 | +19% | $949.4B | $72.51 | | | |

## Balance sheet and share count reconciliation

Enterprise value converts to equity value after net debt, and targets divide by diluted shares.

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| SCENARIO | ENTERPRISE VALUE | EXIT EV/ FCF | TERMINAL GROWTH | PLUS: NET CASH | EQUITY VALUE | TARGET |
|---|---|---|---|---|---|---|
| **Bear** | $809.12B | — | 2.0% | -$56.90B | $752.22B | **$61.12** |
| **Base** | $1,672.72B | — | 2.5% | -$56.90B | $1,615.82B | **$131.29** |
| **Bull** | $3,065.41B | — | 3.0% | -$56.90B | $3,008.51B | **$244.45** |

## Fair value across scenarios

Model fair value by year under each scenario, on the same paths the ticker page chart draws.

| SCENARIO | 2026 | 2027 | 2028 | 2029 | 2030 | 2031 |
|---|---|---|---|---|---|---|
| **Bear** | $61.12 | $66.90 | $76.25 | $85.49 | $93.35 | $98.13 |
| **Base** | $131.29 | $143.70 | $160.42 | $176.71 | $190.74 | $201.38 |
| **Bull** | $244.45 | $267.55 | $293.91 | $319.88 | $343.37 | $362.62 |

Methodology note. The discount rate is the live U.S. 10-year Treasury yield plus beta times equity risk premium. Terminal growth reflects long-run nominal GDP. The $142.04 headline target applies 25% / 50% / 25% Bear / Base / Bull weights.

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Alphabet Inc. valuation framework

# The franchise deep dive

Google Cloud hypergrowth: +82% YoY revenue, tripled operating income, $514B contracted backlog converting over 24 months.

Gemini scale: 950M MAU, 22B API tokens per minute, deepening enterprise AI adoption on Google infrastructure.

TPU monetization: first external sales of complete TPU systems, a new hardware revenue stream from 2027.

Capital return: Berkshire Hathaway's $10B endorsement; ongoing buybacks steadily reduce share count (12.9B to 12.3B diluted since the split).

## Revenue and margin trend

| METRIC | FY2022 | FY2023 | FY2024 | FY2025 | H1 2026 |
|---|---|---|---|---|---|
| **Revenue ($B)** | 282.8 | 307.4 | 350.0 | 402.8 | 229.7 |
| **Operating income ($B)** | 74.8 | 84.3 | 112.4 | 129.0 | 80.5 |
| **Operating margin** | 26.5% | 27.4% | 32.1% | 32.0% | 35.0% |
| **Free cash flow ($B)** | 60.0 | 69.5 | 72.8 | 73.3 | N/A |
| **Capex ($B)** | 31.5 | 32.3 | 52.5 | 91.4 | 80.6 |

## Valuation cross-check

DCF base $131.29 vs $340.27 (-61%). Blended (25/50/25) $142.04. Even the $244.45 Bull case is 28% below the price. 8.9% WACC (live CAPM), 2.5% terminal growth.

At ~8.5x FY2026E revenue and 22.8x forward earnings, GOOGL trades at a premium to slower-growing mega-caps but in line with AI-infrastructure peers. The premium is unsupported by near-term FCF, which is negative under the capex load.

Historically GOOGL traded at 18-25x earnings with 20%+ FCF margins. The current multiple embeds flawless AI-capex ROI; any guide-down in Cloud growth or capex discipline re-rates the multiple lower, as the September 3.8% selloff previewed.

## What to monitor

**~Oct 29, 2026.** Q3 2026 earnings: watch Search and Cloud growth, capex run-rate, and FCF trajectory.

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**Q4 2026.** 2027 capex budget signal: a below-$200B guide with Cloud margins intact would be the single biggest positive re-rating event.

**2027.** TPU system sales ramp to external customers, a new hardware revenue stream.

**Ongoing.** Waymo city expansion and international launches following the $16B round at a $126B valuation.

## Financial summary: income statement

| YEAR | REVENUE | GROSS PROFIT | OPERATING INCOME | EBITDA |
|---|---|---|---|---|
| **2021** | $257.6B | — | $78.7B | — |
| **2022** | $282.8B | $156.6B | $74.8B | $85.2B |
| **2023** | $307.4B | $174.1B | $84.3B | $98.0B |
| **2024** | $350.0B | $203.7B | $112.4B | $135.4B |
| **2025** | $402.8B | $240.3B | $129.0B | $180.7B |

## Balance sheet

| YEAR | TOTAL ASSETS | TOTAL LIABILITIES | TOTAL EQUITY | CASH |
|---|---|---|---|---|
| **2021** | $359.3B | $107.6B | $251.6B | $20.9B |
| **2022** | $365.3B | — | $256.1B | $21.9B |
| **2023** | $402.4B | — | $283.4B | $24.0B |
| **2024** | $450.3B | — | $325.1B | $23.5B |
| **2025** | $595.3B | — | $415.3B | $30.7B |

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## Cash flow

| YEAR | OPERATING CF | CAPEX | FINANCING CF | FREE CF |
|---|---|---|---|---|
| **2021** | $91.7B | $-24.6B | $-61.4B | $67.0B |
| **2022** | $91.5B | $-31.5B | $-69.8B | $60.0B |
| **2023** | $101.7B | $-32.3B | $-72.1B | $69.5B |
| **2024** | $125.3B | $-52.5B | $-79.7B | $72.8B |
| **2025** | $164.7B | $-91.4B | $-37.4B | $73.3B |

Methodology note. The discount rate is the live U.S. 10-year Treasury yield plus beta times equity risk premium. Terminal growth reflects long-run nominal GDP. The $142.04 headline target applies 25% / 50% / 25% Bear / Base / Bull weights.

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# Agency risk score 52 of 100: balanced, typical corporate alignment

The score follows the website's Executive Alignment & Agency Risk model: insider ownership up to 40 points, open-market Form 4 flow over the trailing six months up to 30, and pay-for-performance up to 30. Insiders hold 1.60% of shares outstanding. Form 4 filings over the trailing six months show $0 of open-market purchases against $0 of sales, a net flow of $0. One-year total shareholder return is +35.2% against CEO total pay of $10.91M (Mr. Sundar Pichai), which the model scores as PERFORMANCE EARNED.

**52**
AGENCY RISK SCORE, OF 100

## The three components

| COMPONENT | SCORE | WHAT THE SITE SHOWS |
|---|---|---|
| **Insider Ownership (Skin in the Game)** | 8 / 40 | 1.60% of shares held by insiders |
| **Form 4 Flow (Conviction Flow)** | 18 / 30 | Buys $0 (0) vs sales $0 (0), net $0, trailing six months |
| **Pay-for-Performance** | 26 / 30 | PERFORMANCE EARNED: Stock Outperformance Matches Comp Structure. 1Y TSR +35.2%; Mr. Sundar Pichai, CEO & Director, total pay $10.91M |
| **Agency Risk Score** | **52 / 100** | Regime: BALANCED / TYPICAL CORPORATE |

## SEC proxy audit

N/A. The weekly DEF 14A proxy pipeline holds no CD&A audit record for this ticker (foreign private issuers file no DEF 14A), so no audit findings are shown.

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Methodology. This module reuses the website's Executive Alignment & Agency Risk model unchanged. Scores of 75 and above band as founder-mode, high conviction; 45 to 74 as balanced, typical corporate; below 45 as high agency risk. Inputs are the same feeds behind the site model (Yahoo Finance insider holdings, Form 4 transactions and executive compensation) plus the SEC DEF 14A CD&A audit from the weekly proxy pipeline. Computed 21 October 2026 for this note.

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## The bear case

### AI capex ROI shortfall

$195-205B of 2026 capex (up ~119% YoY), rising again in 2027, with Q2 FCF already negative. If Cloud and AI revenue do not compound fast enough, shareholders fund a multi-year build with no return.

### Search disruption

Meta's Muse agent bypasses the search results page where Google's ad auction operates; 2.8M downloads in twelve days triggered a 3.8% selloff. Agentic AI is the first structural threat to Search monetization.

### Regulatory remedies

Ad-tech rulings require business changes and the search case forces data sharing with competitors plus limits on exclusive distribution deals, chipping at structural advantages.

### Valuation

The market prices a ~63% five-year revenue CAGR per our reverse DCF. Any growth disappointment against that bar implies sharp multiple compression.

### YouTube competition

Q2 YouTube ad revenue missed estimates on competition and shifting marketing budgets, removing a cushion that usually offsets Cloud investment pressure.

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## The strengths

### Operating Quality

One of the highest-quality businesses in the market: 32-36% operating margins, $129B FY2025 operating income, and a fortress balance sheet (now levering deliberately for the AI build, $112.8B debt vs $242.5B cash plus securities). ROIC rema

### Recent Trend

Revenue accelerating: Q1 2026 +21.8%, Q2 2026 +24.4% YoY; Google Cloud +82% YoY; operating margin expanded to 34-36%. Offset by Q2 free cash flow turning negative ($5.9B) as capex doubled, and the stock's 3.8% September selloff on AI-agent

### Forward Growth

Google Cloud carries a $514B backlog (nearly doubled in twelve months) with tripled operating income; Gemini app at 950M MAU processing 22B tokens per minute; TPU system sales to external customers beginning; Waymo valued at $126B after a $

### Valuation

Our DCF base target is $131.29 vs $340.27 (-61%); even the Bull case ($244.45) sits 28% below the price. The reverse DCF shows the market pricing a ~63% five-year revenue CAGR. At 8.5x FY2026E revenue with FCF under acute capex pressure, th

## Cascadia scorecard

**Composite score 64: Fair.** Confidence High, as of 2026-10-18.

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| PILLAR | RATIONALE | SCORE |
|---|---|---|
| **Operating Quality** | One of the highest-quality businesses in the market: 32-36% operating margins, $129B FY2025 operating income, and a fortress balance sheet (now levering deliberately for the AI build, $112.8B debt vs $242.5B cash plus securities). ROIC remains elite on the ad franchise. | 88 |
| **Recent Trend** | Revenue accelerating: Q1 2026 +21.8%, Q2 2026 +24.4% YoY; Google Cloud +82% YoY; operating margin expanded to 34-36%. Offset by Q2 free cash flow turning negative ($5.9B) as capex doubled, and the stock's 3.8% September selloff on AI-agent disruption fears. | 72 |
| **Forward Growth** | Google Cloud carries a $514B backlog (nearly doubled in twelve months) with tripled operating income; Gemini app at 950M MAU processing 22B tokens per minute; TPU system sales to external customers beginning; Waymo valued at $126B after a $16B round. Multiple compounding growth engines. | 85 |
| **Valuation** | Our DCF base target is $131.29 vs $340.27 (-61%); even the Bull case ($244.45) sits 28% below the price. The reverse DCF shows the market pricing a ~63% five-year revenue CAGR. At 8.5x FY2026E revenue with FCF under acute capex pressure, the multiple discounts perfection. | 25 |
| **Catalysts & Risk** | Near-term catalysts: Q3 2026 earnings (~Oct 29), 2027 capex budget signal, Waymo expansion. But the risk ledger is heavy: a $200B+ annual capex supercycle with uncertain returns, Meta's Muse agent bypassing the search results page, ad-tech and search antitrust remedies, and YouTube share pressure. | 55 |

## Investment view

A credible 2027 capex normalization (below $200B with Cloud margins holding), Search monetization re-accelerating past AI-agent fears, or a 10Y yield decline cutting our ~8.9% WACC would lift the DCF materially. Conversely, another capex guide raise, Cloud deceleration, or Search share loss would cut it further.

**Investment view: Hold, Valuation Concerns. The $142.04 blended target implies -58%, balancing the $61.12 / $131.29 / $244.45 Bear / Base / Bull outcomes.**

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